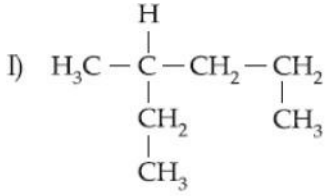
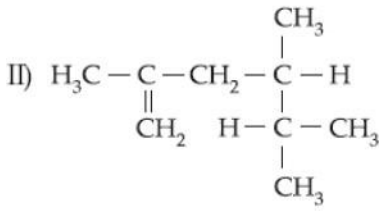
LISTA DE EXERCÍCIOS
NOMENCLATURA DE HIDROCARBONETOS

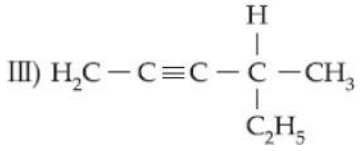
ESQUEMA DE NOMENCLATURA

1º Identificar a cadeia principal;
2º Identificar as ramificações;
3º Numerar a cadeia principal, obedecendo:
1º) Grupo funcional (álcool, cetona, aldeído, ...)
2º) Insaturação (dupla / tripla)
3º) Ramificação
4º Dar o nome das ramificações e depois na cadeia principal.

1. **(Vunesp-SP)** Os nomes corretos dos compostos orgânicos de acordo com a IUPAC:

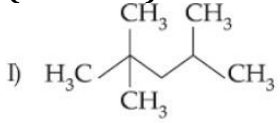


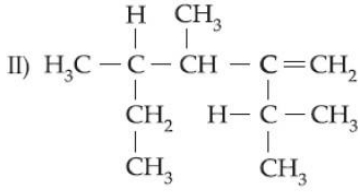


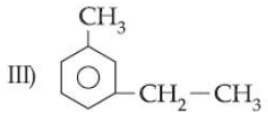


2. **(Mackenzie-SP)** Retirando-se um dos hidrogênios do metano e colocando-se o radical terc-butil, tem-se a fórmula do:
a) dimetil-propano. b) 2-metil-butano.
c) pentano. d) 2,3-dimetil-butano.
e) butano.

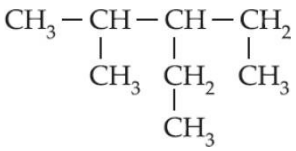
3. **(UECE-CE)** Dê os nomes nos compostos:







4. **(PUC-RS)** O composto orgânico representado a seguir:

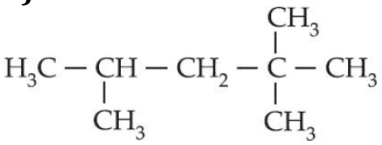


é denominado:
a) 3-etil-2,4-dimetilbutano.
b) 2-etil-1,3-dimetilbutano.
c) 3-isopropilpentano.
d) 3-etil-4-metilpentano.
e) 3-etil-2-metilpentano.

5. **(UEL-PR)** Considere o alcano com o menor número de átomos de carbono apresentando um radical etil ligado à cadeia carbônica principal. Qual o número total de átomos de carbono na molécula do hidrocarboneto?

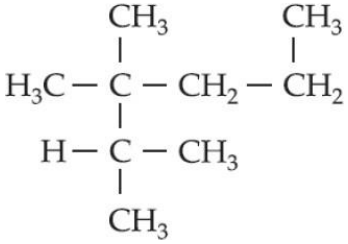
a) 6 b) 7 c) 8 d) 9 e) 10

6. **(PUC-MG)** Dada a cadeia carbônica



verifica-se que a soma das posições dos carbonos na numeração da cadeia é:
a) 4 b) 6 c) 8 d) 10 e) 12

7. **(Vunesp-SP)** O nome correto do composto orgânico de fórmula



a) 3-isopropil-2-metilpentano.
b) 2-isopropil-2,4-dimetilbutano.
c) 2,3,3-trimetilexano.
d) 2,2,4-trimetilpentano.
e) 3,3-dimetil-5-metilpentano.

8. Dê o nome segundo as recomendações da IUPAC para os hidrocarbonetos apresentados abaixo.

a) $CH_3-CH_2-CH_3$

b) CH_3-CH_3

c) $CH_3-CH_2-CH_2-CH_2-CH_3$

d) $CH_2=CH-CH_3$

e) $CH_3-CH_2-CH_3-CH_2-CH_2-CH_2-CH_2-CH_3$

f) $CH_3-CH_2-C\equiv CH$

g) $CH_2=CH-CH=CH-CH_3$

h) $CH_2=CH-CH=CH-CH=CH-CH_3$

i) 

j) 

k) 

l) 

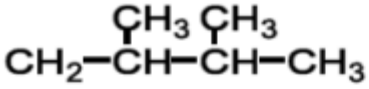
m) 

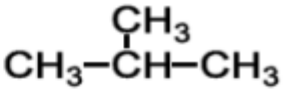
n) 

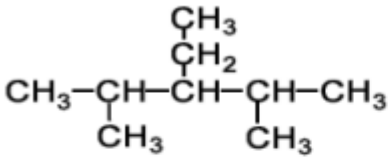
o) 

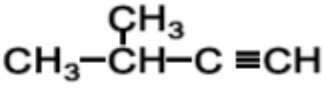
p) 

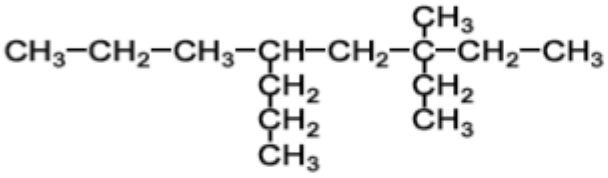
9. Dê o nome segundo as recomendações da IUPAC para os hidrocarbonetos de cadeia ramificada apresentados abaixo.

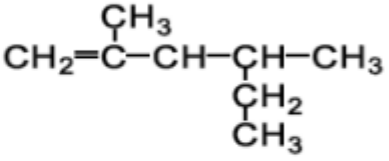
a) 

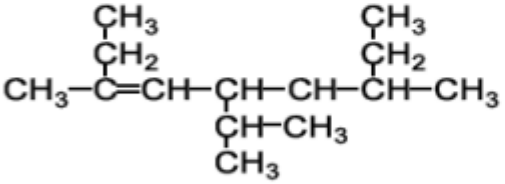
b) 

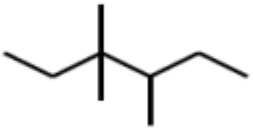
c) 

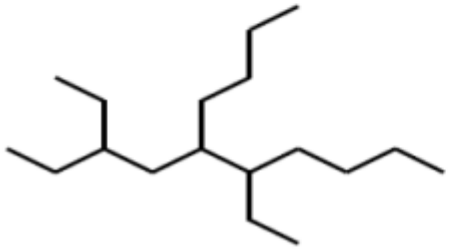
d) 

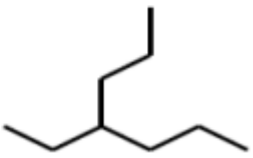
e) 

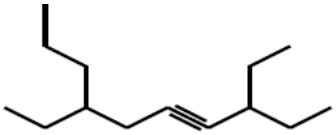
f) 

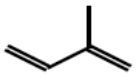
g) 

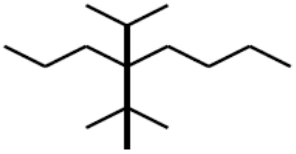
h) 

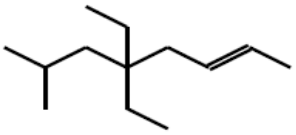
i) 

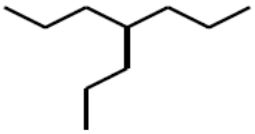
j) 

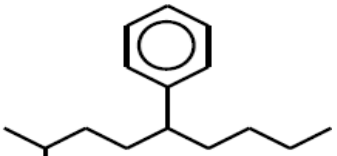
k) 

l) 

m) 

n) 

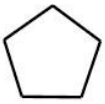
o) 

p) 

10. Dê o nome segundo as recomendações da IUPAC para os hidrocarbonetos cíclicos apresentados abaixo.

a) 

b) 

c) 

d) 

e) 

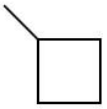
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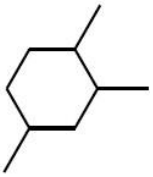
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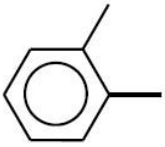
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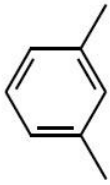
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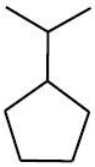
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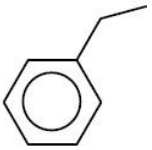
k) 

l) 

m) 

n) 

o) 

p) 

11. Forneça as estruturas dos seguintes hidrocarbonetos:

a) Butano
b) Heptano
c) Pent-2-ino
d) Propino
e) Eteno
f) Nona-1,3,6-trieno
g) 3-metilpentano
h) 2,2-dimetilbutano
i) 3-etil-2,5,5-trimetiloctano
j) 5-sec-butil-deca-2,6-dieno
k) para-dimetilbenzeno
l) meta-dietilbenzeno
m) 3-etil-ciclo-hexeno
n) 1,1,2-trimetilciclopentano

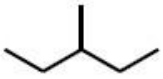
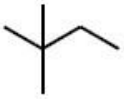
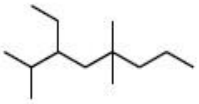
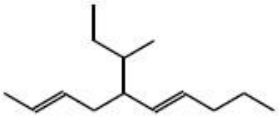
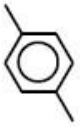
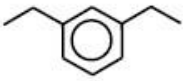
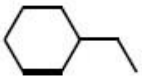
GABARITO

01- I) 3-metil hexano ; II) 2,4,5 trimetil hex-1-eno ; III) 4-metil hex-2-ino ; 02- A ; 03- I) 2,2,4-trimetil-pentano (isoctano) ; II) 2-isopropil- 3,4-dimetil hex-1-eno ; III) 1-etil-3-metil-benzeno ; 04- E; 05- B ; 06- C ; 07- C

08-
a) propano
b) etano
c) pentano
d) propeno
e) octano
f) but-1-ino
g) penta-1,3-dieno
h) hepta-1,3,5-trieno
i) hexano
j) but-2-ino
k) decano
l) hepta-2,4-dieno
m) nona-3,5-diino
n) oct-4-ino
o) nonano
p) oct-2-eno

9-
a) 2,3-dimetilbutano
b) metilpropano
c) 3-etil-2,4-dimetilpentano
d) 3-metilbut-1-ino
e) 3-etil-3-metil-5-propiloctano
f) 2,4-dimetil-hex-1-eno
g) 5-isopropil-3,7-dimetilnon-3-eno
h) 3,3,4-trimetil-hexano
i) 5-butil- 3,6-dietildecano
j) 4-etil-heptano
k) 3,7-dietildec-4-ino
l) 2-metilbuta-1,3-dieno
m) 4-isopropil-4-terc-butiloctano
n) 5,5-dietil-7-metiloct-2-eno
o) 4-propil-heptano
p) 5-fenil-2-metilnonano

10-
a) ciclopropano
b) ciclobutano
c) ciclpentano
d) ciclo-hexano
e) ciclooctano
f) benzeno
g) ciclobuteno
h) ciclo-hexeno
i) ciclo-penta-1,3-dieno
j) 1-metilciclo-hex-1-eno
k) metilciclobutano
l) 1,2,4-trimetilciclo-hexano
m) 1,2-dimetilbenzeno
m) 1,3-dimetilbenzeno
o) isopropilciclopentano
p) etilbenzeno

11-
a) 
b) 
c) 
d) 
e)
f) 
g) 
h) 
i) 
j) 
k) 
l) 
m) 
n) 